Engineering - Undergraduate Level Assessment 5 Total Marks: 40 Answer all questions.

# Multiple Choice (10 marks)

1. Find the inverse Laplace transform L^-1[1 / {s(s^2 + 1)}], using the convolution.
   (a) sin t - t
   (b) 1 - cos t
   (c) t sin t
   (d) e^t - 1
   (e) 1 - e^-t
2. Which of the following is not a sinusoidal oscillator?
   (a) Hartley oscillator
   (b) Colpitts oscillator
   (c) Clapp oscillator
   (d) Quartz crystal oscillator
   (e) LC oscillator
3. Lowest critical frequency is due to zero and it may be present at the origin or nearer to the origin, then the type of network is
   (a) RL Circuit.
   (b) RC circuit.
   (c) CL circuit.
   (d) CR circuit.
   (e) RCL circuit.
4. Find the Laplace transform of f(t) = t^n, where n is a positive integer.
   (a) n^n / s^n
   (b) n! / s^n+1
   (c) n! / s^n+2
   (d) (n+1)! / s^n
   (e) n! / s^n
5. A single phase full bridge inverter can operated in load commutation mode in case load consist of
   (a) Purely resistive load.
   (b) RL.

(c) RLC underdamped with a parallel connection.

(d) RLC critically damped.

(e) RLC underdamped with a series connection.

## True / False (10 marks)

*Answer True or False*

6. True or False: In ferromagnetic materials, the magnetisation and applied field are related linearly

7. True or False: The attenuation per meter along the wave guide for a wavelength of 2 meters is 54

8. True or False: In a phase-shift keying system with imperfect synchronization, a phase shift of 24 = P_e(DPSK) = 10^-5.

9. True or False: The solid angle subtended by an area of 2400 cm^2 on a sphere with a diameter o

10. True or False: The initial current i(0^+) can be expressed as E / (v(0^+) * R).

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the change in enthalpy and entropy for a real gas along an isothermal path between pressures P_1 and P_2, using the provided equation for v and the constants a and b.

12. Discuss the back and front pitches of a 4-pole machine with lap and wave windings, considering the implications of having 36 and 37 winding elements, respectively.

*End of Paper*